

Wild Multiplicity, Exact Residue Degrees and the Geometric Zeta of the Additive Map $x \mapsto x + x^p$

HCS-C384 / HEN-O368

5 September 2026

Abstract

For every prime characteristic, including two, we distinguish the geometric fixed points of $x \mapsto x + x^p$ from the lengths of its fixed schemes. At iterate n , writing $e = p^{v_p(n)}$, the scheme has length p^n but only p^{n-e} geometric points, each of multiplicity p^e . This identifies the exact wild correction and all primitive-cycle counts. A linearized-polynomial Euclidean argument gives every finite-extension count without assuming that either clock is prime to the characteristic. We determine the least splitting degree and separate dynamical least period from residue degree by a double Möbius inversion. A valuation expansion yields a Lambert series with an explicit interior tail bound. At every nontrivial root of unity of p -power order its radial logarithmic-derivative residue is strictly between zero and one, ruling out local meromorphic continuation of the geometric zeta itself. Thus its circle $|z| = 1/p$ is a meromorphic natural boundary, although length counting gives the rational function $(1 - pz)^{-1}$. Independent exact computations validate 1,344 iterate cells and 5,760 extension cells. The result is a source-specific theorem synthesis and audit, not a new Riemann determinant or a claim of priority over the classical work of Bridy.

Keywords: additive polynomials; wild multiplicity; periodic points; finite fields; dynamical zeta; natural boundary

中文摘要

本文研究所有素特征中的加性映射，并严格区分几何固定点与概形长度。迭代中的野性重数由素进赋值完全决定，纯素数幂周期退化单独处理。线性化多项式的欧几里得算法给出全部扩域计数，不要求扩域次数与特征互素。双重莫比乌斯反演区分动力学周期和算术剩余域次数。显式朗伯级数及其非整数留数证明几何塞塔函数存在亚纯自然边界。概形长度的有理行列式不能替代真正的周期轨道计数；本文不作目标黎曼函数或文献优先权声明，所有计算仅作为完整解析证明的回归验证。

关键词： 加性多项式；野性重数；周期点；有限域；动力学塞塔；自然边界

1 The source and the precise contribution

Fix a prime p and $f(x) = x + x^p$ on $\mathbb{A}^1(\overline{\mathbf{F}}_p)$. The parameter p indexes source fibres; it is not a period label assigned to a rational prime. The clock is composition number $n \geq 1$. The point at infinity is not adjoined. For a polynomial $Q(T) = \sum q_j T^j$ over $\mathbf{F}_p$, set $L_Q(X) = \sum q_j X^{p^j}$. Direct expansion gives $L_{AB} = L_A \circ L_B$ and $L_{A+B} = L_A + L_B$.

Transcendence for additive-polynomial dynamical zeta functions is classical: Bridy's earlier theorem covers $x^{p^m} + ax$ in odd characteristic [1, Theorem 2], and the later additive-polynomial theorem includes all positive characteristics [2, Theorem 1.3]. We claim no priority for transcendence or natural boundaries. Our contribution within this research sequence is a complete, source-locked account of geometric versus scheme counting, with independent evidence and explicit restrictions on determinant promotion. The all-parameter proof does not depend on the finite computational grid.

2 All iterates and their wild multiplicities

Round 0 advance. The geometric census and the scheme-length census differ at every iterate, and the difference has an exact formula.

Theorem 1 (Geometric and scheme ledgers). *Write $n = em$, $e = p^{v_p(n)}$, with $p \nmid m$. The fixed scheme of f^n has length p^n . Its geometric points number $N_n = p^{n-e}$ and each has local multiplicity p^e .*

Proof. In the commutative additive-endomorphism ring, $f = 1 + F$ with $F(X) = X^p$. Thus $f^n - \text{id} = L_{Q_n}$, where

$$Q_n(T) = (1 + T)^n - 1 = T^e H_m(T)^e, \quad H_m(T) = \frac{(1 + T)^m - 1}{T}.$$

Since $H_m(0) = m \neq 0$, the linearized polynomial $L_{H_m^e}$ is separable of degree $p^{e(m-1)} = p^{n-e}$. Moreover

$$f^n(X) - X = L_{H_m^e}(X)^{p^e}.$$

Every root is therefore repeated exactly p^e times. The product of the number of roots and their common multiplicity is p^n , the degree of the original monic polynomial. $\square$

For $n = p^v$ one has $H_1 = 1$, so the only geometric fixed point is 0, and $f^{p^v}(X) - X = X^{p^{p^v}}$. In particular no nonzero point has pure p -power period. The differential of f is 1 everywhere, and every periodic multiplier is 1. Wild scheme multiplicity is not a real Lyapunov exponent or a nondegenerate flat-trace denominator.

On each finite fixed set, f is a permutation because $f^n = \text{id}$ there. Consequently forward-oriented primitive-cycle counts are

$$P_n = \frac{1}{n} \sum_{d|n} \mu(n/d) p^{d - p^{v_p(d)}} \in \mathbf{Z}_{\geq 0}, \quad N_n = \sum_{d|n} d P_d. \quad (1)$$

No reversal quotient is imposed; every geometric orbit has weight one. The scheme-length exponential instead satisfies

$$\zeta_p^{\text{len}}(z) = \exp \left(\sum_{n \geq 1} p^n \frac{z^n}{n} \right) = \frac{1}{1 - pz}. \quad (2)$$

This is a scalar length-counting determinant, not a geometric orbit trace. The discrepancy is already p versus 1 at the first iterate.

3 Every extension and the two-clock dictionary

Round 1 advance. The extension census is closed without discarding repeated factors in characteristic-divisible extension degrees.

Theorem 2 (All extension degrees). *For all $n, r \geq 1$,*

$$N_{n,r} := \#\{x \in \mathbf{F}_{p^r} : f^n(x) = x\} = p^{\deg \gcd((1+T)^n - 1, T^r - 1)}.$$

For $m > 1$, let o_m be the order of the unit T in $\mathbf{F}_p[T]/(H_m)$. The least extension containing all geometric fixed points of f^n has degree eo_m . For $m = 1$ that degree is 1.

Proof. Euclidean division $A = BQ + R$ implies $L_A = L_Q \circ L_B + L_R$. Hence L_A, L_B have the same common roots as L_B, L_R . Iterating the Euclidean algorithm shows that their common roots are the roots of the monic linearized gcd. Apply this with $A = (1 + T)^n - 1$, $B = T^r - 1$; then $L_B = X^{p^r} - X$. The gcd g has nonzero constant coefficient because $T \nmid B$, so L_g is separable and has $p^{\deg g}$ roots. This argument permits $p \mid n$ and $p \mid r$; replacing B by its radical would be an error.

The polynomial H_m is squarefree, since $(1 + T)^m - 1$ has no common root with its derivative $m(1 + T)^{m-1}$. Its constant coefficient is nonzero. Thus the finite algebra $\mathbf{F}_p[T]/(H_m)$ is a product of finite fields, and o_m is prime to p . The gcd attains degree $n - e$ exactly when $H_m^e \mid T^r - 1$. Factoring the latter polynomial into its prime-to- p part with multiplicity $p^{v_p(r)}$ shows this is equivalent to $eo_m \mid r$. For $m = 1$ the root set is $\{0\}$. $\square$

Let $E_{n,r}$ count points of exact dynamical period n and exact residue degree r . A polynomial defined over $\mathbf{F}_p$ cannot increase residue degree. On a periodic cycle the field inclusions return to the starting point, so all inclusions are equalities. Double Möbius inversion gives

$$E_{n,r} = \sum_{a|n} \sum_{b|r} \mu(n/a) \mu(r/b) N_{a,b}, \quad E_{n,r}/n \in \mathbf{Z}_{\geq 0}. \quad (3)$$

The quotient counts f -cycles whose points have exact residue degree r . It does not count joint orbits under f and Frobenius: the latter could identify several f -cycles. Thus no hidden identification of the two clocks is used.

Since $N_n \leq p^n$ and $N_n = p^{n-1}$ when $p \nmid n$, the logarithmic series has radius exactly $1/p$. Within $|z| < 1/p$,

$$\zeta_p(z) = \exp \left(\sum_{n \geq 1} N_n \frac{z^n}{n} \right) = \prod_{n \geq 1} (1 - z^n)^{-P_n}. \quad (4)$$

Indeed $nP_n \leq N_n$ ensures absolute convergence; expanding each logarithm and using (1) justifies the exchange of sums. We call $D_p = 1/\zeta_p$ the native primitive determinant germ, without assuming an operator realization.

4 An explicit meromorphic natural boundary

Round 2 advance. The natural-boundary claim is proved for the zeta itself, not inferred informally from a singular derivative. The argument also supplies a rigorous interior tail bound.

Set $w = pz$, $\mathcal{Z}_p(w) = \zeta_p(w/p)$, and

$$A_p(w) = w \frac{\mathcal{Z}'_p(w)}{\mathcal{Z}_p(w)} = \sum_{n \geq 1} p^{-p^{v_p(n)}} w^n.$$

Define $c_0 = p^{-1}$ and $c_k = p^{-p^k} - p^{-p^{k-1}} < 0$ for $k \geq 1$. The telescoping coefficient identity yields

$$A_p(w) = \sum_{k \geq 0} c_k \frac{w^{p^k}}{1 - w^{p^k}}, \quad |w| < 1. \quad (5)$$

For $|w| \leq \rho < 1$ and $K \geq 0$, its tail is bounded by

$$\left| \sum_{k > K} c_k \frac{w^{p^k}}{1 - w^{p^k}} \right| \leq p^{-p^K} \frac{\rho^{p^{K+1}}}{1 - \rho^{p^{K+1}}}. \quad (6)$$

This follows from $\sum_{k > K} |c_k| = p^{-p^K}$ and monotonicity of $\rho^h/(1 - \rho^h)$. It proves normal convergence on compact subdisks. With each logarithm normalized at zero, integration gives the local formula

$$\log \mathcal{Z}_p(w) = - \sum_{k \geq 0} \frac{c_k}{p^k} \log(1 - w^{p^k}).$$

No branch continuation beyond the unit disk is built into this equality.

Theorem 3 (Natural boundary and HEN-O368). *The circle $|z| = 1/p$ is a meromorphic natural boundary for both ζ_p and D_p . Neither has a same-variable meromorphic global completion, even after multiplication by an entire zero-free function.*

Proof. Fix a root of unity ω of exact order p^K , $K \geq 1$. In (5), the finitely many terms with $k < K$ vanish after multiplication by $1 - r$ and substitution $w = r\omega$, $r \uparrow 1$. For $k \geq K$, $\omega^{p^k} = 1$, and

$$0 \leq (1 - r) \frac{r^{p^k}}{1 - r^{p^k}} \leq 1, \quad (1 - r) \frac{r^{p^k}}{1 - r^{p^k}} \rightarrow p^{-k}.$$

Summability of $|c_k|$ permits dominated convergence. Hence

$$(1 - r)A_p(r\omega) \rightarrow S_K := \sum_{k \geq K} c_k p^{-k} < 0, \quad 0 < -S_K \leq p^{-K-p^{K-1}} < 1.$$

If $\mathcal{Z}_p$ had a meromorphic continuation near ω , it would be nonzero as a germ and its local order j would be an integer. The logarithmic derivative would then have residue j . But its radial limit is

$$\lim_{r \uparrow 1} (r\omega - \omega) \frac{\mathcal{Z}'_p(r\omega)}{\mathcal{Z}_p(r\omega)} = -S_K \in (0, 1),$$

which is impossible. This includes the boundary case $p = 2, K = 1$. Nontrivial p -power roots are dense on the unit circle. Any continuation through another point would define a meromorphic germ at one of these roots and agree on the connected overlap inside the disk, a contradiction. The reciprocal has the same obstruction because reciprocation preserves nonzero meromorphic germs. Multiplication by an entire zero-free factor cannot help: division by that factor would restore a forbidden continuation. $\square$

The theorem does not exclude an operator determinant defined only inside the original disk, distributional boundary values, a changed variable with a different domain, or a genuinely different dynamical object. It does exclude replacing the geometric ledger by (2) and claiming that the original primitive determinant thereby continues.

5 Independent exact evidence and source collisions

The producer implements modular Euclid and direct finite-field arithmetic. The checker imports no producer: it uses sparse polynomial Euclid and a matrix-rank recount of the nine small fields. SymPy separately checks binomial factorization, extension gcds and the first sixteen primitive-product coefficients. All computations are exact; no numerical target zeros are used.

Frozen regression population	Size
Prime-iterate rows: 14 primes through 43, $1 \leq n \leq 96$	1,344
Extension rows: $p = 2, 3, 5, 7, 11$, $n \leq 48$, $r \leq 24$	5,760
Neighboring coefficient rows: $a \in \mathbf{F}_p$, $n \leq 32$	896
Direct field recounts: $p = 2, 3, 5$, $r = 1, 2, 3$	9
Rational noninteger-residue enclosures	9
Rational interior-tail certificates	18

Two independent working directories reproduce byte-identical evidence. Hostile tests repair the payload hash before attacking geometric counts, multiplicities, characteristic-divisible extensions, primitive counts, residue signs, source clocks, schema and target flags. These are execution and schema checks, not machine proofs of the infinite theorem.

Repository C204 already closes a fixed finite-dimensional finite-field linear system. C14 studies chronological S -integer solenoid zeta, C159 a Thue–Morse renewal natural boundary, and C382 reduced elliptic Frobenius fixed points. The current infinite geometric population, wild fixed-scheme multiplicity and complete extension dictionary are a different owner. The Bridy results remain direct prior art, not negative controls to be ignored.

6 Controls and the target boundary

The parent $g_0(x) = x^p$ has $(g_0^n - x)' = -1$ and p^n distinct fixed points, so its length and geometric ledgers agree. For $g_a(x) = ax + x^p$, $a \in \mathbf{F}_p^\times$ of order d , the binomial polynomial $(a + T)^n - 1$ has nonzero constant term if $d \nmid n$; otherwise its lowest nonzero term has degree $p^{v_p(n)}$. Thus the counts are respectively p^n and $p^{n-p^{v_p(n)}}$. This is an intrinsic derivative/unit-order control. Mixed composite integers cannot be field characteristics; prime powers specify extension sizes over a fixed prime.

The conservative evaluator result is

$$(A0_STRUCTURAL_ARITHMETIC_RELATION, A1_WEAK, A2_FAIL, A3_FAIL, A4_FAIL).$$

Finite-field arithmetic is present, but no autonomous primitive orbit is assigned to each rational prime and no target log-prime period law follows. The six mandatory A1 controls are not completed. No target Euler factors, root number, automorphy, target divisor, target functional equation, target zero match or Hilbert–Pólya operator is claimed.

Route B remains disabled.

Reproducibility and research-process disclosure

This work used AI-assisted mathematical drafting and exact-code development. The proof, source audit, actual same-family cross-agent review, independent checker, symbolic lane, repaired-hash attacks and deterministic compilation receipts are retained. Separate-agent review exposed and corrected a standalone-checker type-coercion gap: prime and extension labels now reject boolean and floating-point aliases. The revised hostile suite rejects 58 of 58 attacks. Independent implementations and separate-agent reviews reduce particular failure risks but are not independent-model or formal-verification certificates. No unpublished material was uploaded to an external model, and no literature-first claim is made.

References

- [1] A. Bridy, Transcendence of the Artin–Mazur zeta function for polynomial maps of $\mathbb{A}^1(\overline{\mathbf{F}}_p)$, *Acta Arithmetica* **156** (2012), 293–300. Author preprint: <https://arxiv.org/abs/1202.0362>.
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